

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWER SCHEME

CLASS

INDEX NUMBER

BIOLOGY

9744/04

Paper 4 Practical

17 August 2017

2 hours 30 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided in the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	24
2	17
3	14
Total	55

This document consists of **13** printed pages and **1** blank page.



Answer **all** questions

- 1 The enzyme **E** catalyses the hydrolysis of sucrose to produce fructose and glucose.

The products of the hydrolysis of sucrose will reduce potassium permanganate from purple to colourless as follows:

purple $\longrightarrow$ colourless

You are required to investigate the effect of sucrose concentration on the progress of this enzyme-catalysed reaction by finding the time taken for the decolourization of potassium permanganate.

You are provided with

- 1% enzyme solution, labelled **E**
- 10% sucrose solution, labelled **S1**
- 2%, 4%, 6%, 8%, 10% glucose solution, labelled **G1 – G5**
- 1 mol dm⁻³ sulfuric acid, labelled **A**
- 0.01% potassium permanganate solution, labelled **P**
- distilled water labelled, **W**

 **Sulfuric acid and potassium permanganate are harmful. If any comes into contact with your skin wash immediately under cold water. It is recommended that you wear safety goggles.**

Proceed as follows:

- 1 Prepare an appropriate volume of 5% sucrose and label it **S2**.
- 2 Put 1 cm³ of **A** into a test-tube.
- 3 Add 1 drop of **P** into the same test-tube. Gently shake to mix.
- 4 Add 1 cm³ of **G1** to the test-tube. Start the stopwatch.
- 5 Record the time taken for **P** to decolourise in step **12**.
- 6 Repeat step **2** to **5** for **G2** to **G5**. You may perform the test simultaneously.
- 7 Put 5 cm³ of **S1** into a small beaker.
- 8 Add 1 cm³ of **E** into the small beaker containing **S1**.
- 9 Stir to mix the solutions. Allow the reaction to take place for 2 minutes.
- 10 Perform step **2** to **5** for reaction mixture of **E** and **S1**.
- 11 Repeat step **7** to **10** for **S2** you have prepared in step **1**.

- 12 Record your data in a suitable format in the space provided below. If **P** does not decolourise, record 'more than 600'.

1. **[L] layout** ®if rate is calculated
2. **[H] headings with units**
3. **[G] time taken for P to decolourise for glucose standards in seconds**
4. **[T] with correct trend (decreasing timing with increasing conc.)**
5. **[S] time taken for P to decolourise for S1 and S2 in seconds, S1 < S2**
®if S1 > S2

Solution	Reducing sugar conc. / %	Time taken for P to decolourise / s		
		CBL	GAR	WMY
G1	2	180	120	210
G2	4	130	150	120
G3	6	110	120	105
G4	8	95	120	80
G5	10	90	75	60
S1	unknown	16	15	20
S2	unknown	20	18	30

-1 for table in pencil. Table must be in black or blue ink.
® time taken

[5]

- (a) (i) Using your results in step 12, estimate the concentration of reducing sugar in the reaction mixture with

S1 corresponding range

[1]

S2 in %

[1]

@ ranges between 2 – 10%

® value between 2 – 10%

® values smaller than 2 and greater than 10

® if missing units (%)

- (ii) With reference to enzyme action, explain your observations for S1 and S2.

1. S1 with higher [S]

thus more ES cplx formed

2. higher rate of sucrose hydrolysed to monosacc / RS / glucose & fructose

thus faster rate of reduction of KMnO_4 , lesser time taken for colour change

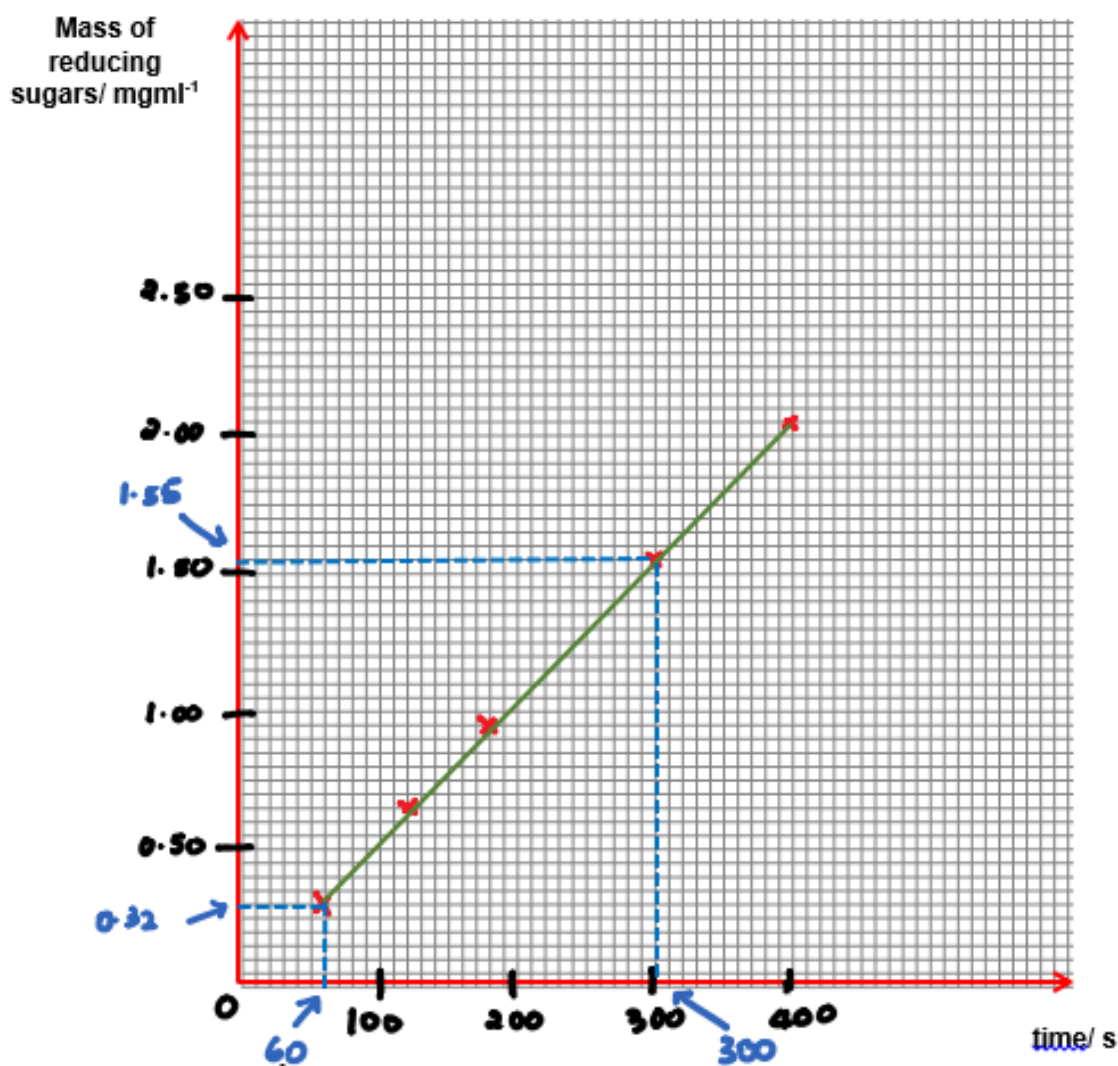
[2]

- (b) Describe a suitable control for this investigation.
 1. *replace E with 1 cm³ of boiled and cooled E / distilled water*
@ replace S with 5cm³ of distilled water
 2. *keeping the rest of the conditions the same as experimental set up*
®point 2 not awarded if point 1 is incorrect [2]
- (c) (i) Identify **two** significant sources of error in this investigation.
 1. *determination of end-point is subjective*
 2. *temp of enzyme reaction not kept constant* [2]
- (d) Table 1.1 shows the results for a similar investigation which measured the mass of reducing sugars produced over a period of 400 seconds.

Table 1.1

time / s	mass of reducing sugars / mg ml ⁻¹
60	0.32
120	0.64
180	0.95
300	1.55
400	2.05

- (i) Plot a graph of the data shown in Table 1.1.



1. [A] correct axes (y-axis: mass of reducing sugars, x-axis: time)
2. [U] correct axes labels with units (y-axis: mg ml^{-1} , x-axis: s)
3. [S] appropriate scale ($> \frac{1}{2}$ total no. of grids provided) AND correct data points plotted
Also, need to show regular intervals on both x-axis & y-axis
4. [L] best fit line or dot-to-dot plot AND no extrapolation (no extension of line graph beyond 1st & last point)

[4]

(ii) Using your graph, find the rate of hydrolysis of the sucrose.

Need to show on graph how gradient is obtained from 2 points on the line graph (with dotted lines extended to scale on y-axis & x-axis shown)

Show on your graph where you took the readings to calculate the rate.
Show all working in your calculation.

[1]

1. calculates gradient (advisable to use 2 points on line graph)(show working)
2. state rate in $\text{mg ml}^{-1} \text{ s}^{-1}$ to 3 s.f.

rate of enzyme activity [1]

- (iii) Describe how results shown in Table 1.1 can be obtained if you are provided with Benedict's solution, 3 mg ml⁻¹ reducing sugar solution and a colorimeter.

1. dilute 3 mg ml⁻¹ reducing sugar (RS) solution with water

to obtain RS solutions of 1, 1.5, 2, 2.5 mg ml⁻¹ as colour glucose standards;;

2. conduct Benedict's test using 2 cm³ of 1, 1.5, 2, 2.5 and 3 mg ml⁻¹ RS

with equal vol of Benedict's soln, place in boiling water bath for 2 min;;

3. record absorbance by ppt formed using a colorimeter

plot a graph of absorbance agst RS conc.;;

4. conduct Benedict's test on 2 cm³ of reaction mixture at 60, 120, 180, 300, 400s

using 2 cm³ of Benedict's soln;;

5. record absorbance using colorimeter

locate RS conc. corresponding to absorbance at each sampling times;;

[5]

[Total: 24]

- 2 K1** is a stained, longitudinal section of a young root tip.

Use your microscope to examine carefully the regions labelled **X** and **Y** in Fig. 2.1.

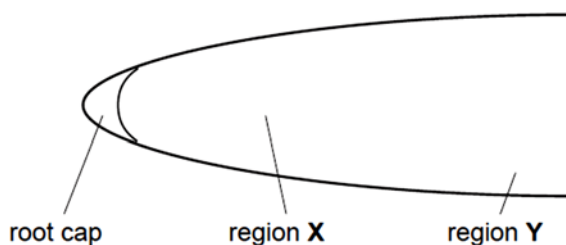


Fig 2.1

- (a)** Make a large, labelled, high-power drawing of a single cell at

- (i)** metaphase

1. [N] draws correct no. of cell with clean, continuous lines

2. [S] of appropriate shape

3. [CW] with thin (proportional) cell wall

4. [C] appropriate chromosomal arrangement

(any 2 MP1-4 for 1m)

5. [L1][L2] with at least 2 labels

cell wall, cytoplasm, cell surface memb

Magnification =

[3]

(ii) anaphase

1. **[N] draws correct no. of cell with clean, continuous lines**
2. **[S] of appropriate shape**
3. **[CW] with thin (proportional) cell wall**
4. **[C] appropriate chromosomal arrangement**
- (any 2 MP1-4 for 1m)
5. **[L1][L2] with at least 2 labels**
cell wall, cytoplasm, cell surface memb

Magnification =

[3]

- (b) (i) Using the eyepiece graticule fitted in the eyepiece lens of your microscope, state the objective lens you are using and the number of eye piece divisions equivalent to length of the cell you have drawn in **(a)(i)** and **(ii)**.

objective lens	x40	x60	
number of eyepiece graticule divisions for (a)(i) =	5 to	8 to	[1]
number of eyepiece graticule divisions for (a)(ii) =	12	20	[1]

- (ii) Using the stage micrometer, determine the length of one division on your eyepiece graticule at the objective stated in **(b)(i)**.

Show the measurements you have made and your working.

1. **100 eyepiece graticule div = 17 stage micrometer div**
2. **$\therefore 1 \text{ eyepiece graticule div} = 0.17 \text{ stage micrometer div}$**
 $= 0.17 \times 0.01 \text{ mm}$
 $= 0.0017 \text{ mm}$
 $= 1.7 \mu\text{m}$

[2]

- (iii) Using your results in **(b)(i)** and **(ii)**, find the actual length, in μm , of the length of the cells that you have drawn in **(a)(i)** and **(ii)**.

Show your working in the space provided.

no. of eye piece graticule div x length of 1 eyepiece graticule division

1. **for (a)(i)**
2. **for (a)(ii)**

(A) ECF from (b)(i) and (ii)

NB: $10 \mu\text{m} < \text{cell size} < 50 \mu\text{m}$

[2]

- (iv) Indicate the actual length of the cells in an appropriate manner on your diagram in (a). [1]
 @ ECF from (b)(iii) i.e. label the wrong actual size but in a correct manner

- (v) Calculate and state the magnification of your drawing in (a). Show your working. [2]
 @ ECF from (b)(iii) i.e. correct calculation using wrong actual size
 ® drawing size measurement of $> \pm 0.2$ cm diff

- (c) Describe **two** differences observed between cells at region X and Y.

1. many cells in region X are undergoing cell division / mitosis

but few / no cells in region Y are dividing

2. cells in region X are small

cells in region Y are larger

3. cells in region X are squarish in shape

cells in region Y are elongated

® cells in region X has no nucleus, cells in region Y has nucleus

[2]

[Total: 17]

- 3 Respiratory quotient (RQ) is a measurement of the ratio of carbon dioxide given out to oxygen taken in. The RQ value act as an indication of the respiratory substrate used.

$$RQ = \frac{\text{CO}_2 \text{ given out}}{\text{O}_2 \text{ taken in}}$$

Carbohydrates often give a RQ of 1.0, while protein and fats give 0.8 and 0.7, respectively.

Yeast are unicellular eukaryotic organisms that respire using a range of substrate

You are required to plan, but not carry out, an experiment to investigate the RQ when yeast is metabolising different carbohydrates in respiration.

You must use:

- active yeast suspension in small conical flask
- 5% glucose
- 5% sucrose
- rubber bung with delivery tube
- soda lime in syringe
- T-shaped connecting tube
- capillary tube
- blue ink

Fig. 3.1 shows part of the experimental setup.

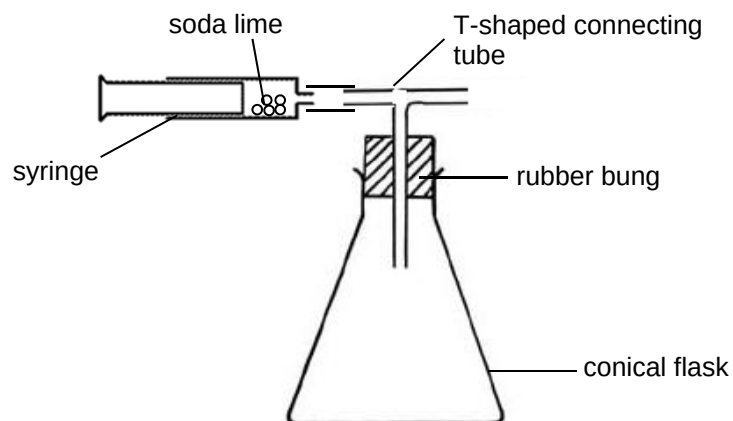


Fig. 3.1

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- thermostatically regulated electrical water bath

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment

[Total: 14]

[T] 1. *explains that O₂ is taken in as final e- acceptor in ETC*

CO₂ is given out in link rxn & Krebs cycle

[IV] 2. *states indep var is type of carbohydrate*

[DV] 3. *states dep var is RQ*

measured by O₂ uptake and CO₂ given out

4. *describes how O₂ uptake is measured e.g. distance moved by ink droplet in presence of soda lime*

explain CO₂ given out is absorbed by soda lime thus Δ in air vol is solely due to O₂ taken in

5. *describes how CO₂ given out is measured e.g. distance moved by ink droplet in absence of soda lime*

explain CO₂ given out is no longer absorbed thus ink movement is due to net diff in O₂ taken in & CO₂ given out

6. *calculate CO₂ given out as difference in distance travelled by ink droplet with & without soda lime*

[CV] 7. *identify controlled var e.g. vol of yeast & resp substrate*

describes how to control it e.g. using 5 cm³ and 10 cm³

8. *identify controlled var e.g. temp*

	<i>describes how to control it e.g. 35°C in thermostatically-regulated electrical water bath</i>
[CS]	9. describe control e.g. replace active yeast with boiled and cooled yeast of same vol, all other conditions same as experimental setup
	<i>state purpose of control</i>
[P]	10. equilibration at desired temp
	<i>acclimatisation after adding resp substrate to yeast</i>
	11. logical, coherent seq of steps
	<i>reasonable duration of rxn e.g. 2 min</i>
	12. diagram of setup e.g. fitting of capillary tube, water bath etc.
[R+R]	13. 3 replicates
	<i>2 repeats</i>
[R]	14. results table with appropriate layout
	<i>headings and units</i>
[S]	15. identify hazard, its corresponding risk and describes safety precaution
	<i>e.g. HCO_3 is skin irritant, wear gloves when handling</i>

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